Structures

Group 8

March 16, 2026

Contents

1 Introduction **1**

2 Deliverable 1: Geometry **2**

3 Deliverable 2: Material, Section, and Loading Data **3**

4 Deliverable 3: Initial Model Validation **4**
4.1 Assumptions 4
4.2 Model Validation 4
4.3 Model Adequacy 4

5 Deliverable 4: Mesh Refinement Sensitivity Analysis **5**
5.1 Introduction and Methodology 5
5.2 Mesh Definitions 5
5.3 Comparison of Key Response Quantities 5
5.4 Conclusion and Adopted Mesh Density 5

6 Deliverable 5: Influence of Prestressing Assumptions **6**
6.1 Introduction and Modelling Basis 6
6.2 Comparison of Key Response Quantities 6
6.3 Discussion and Implications 6

7 Deliverable 6: Boundary Conditions for More Detailed Analysis **7**

1 Introduction

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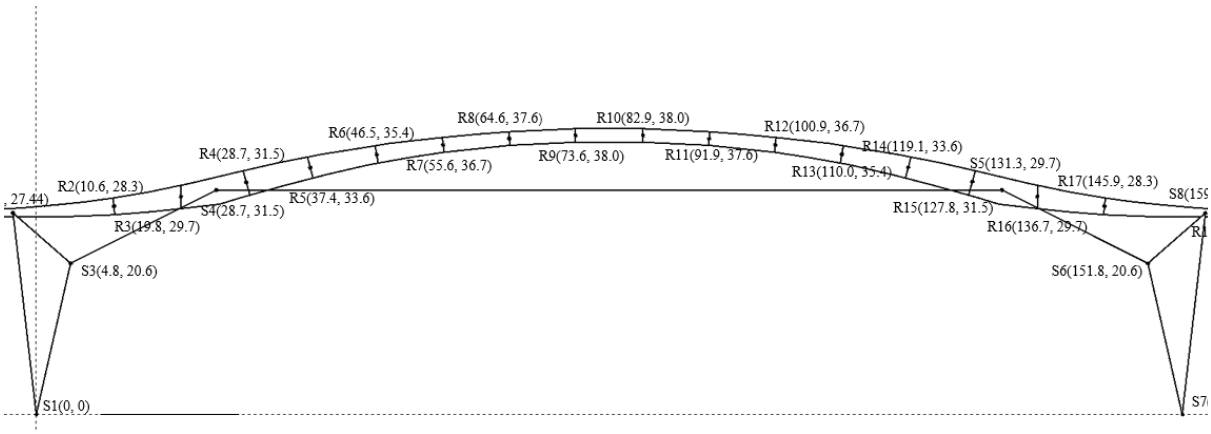
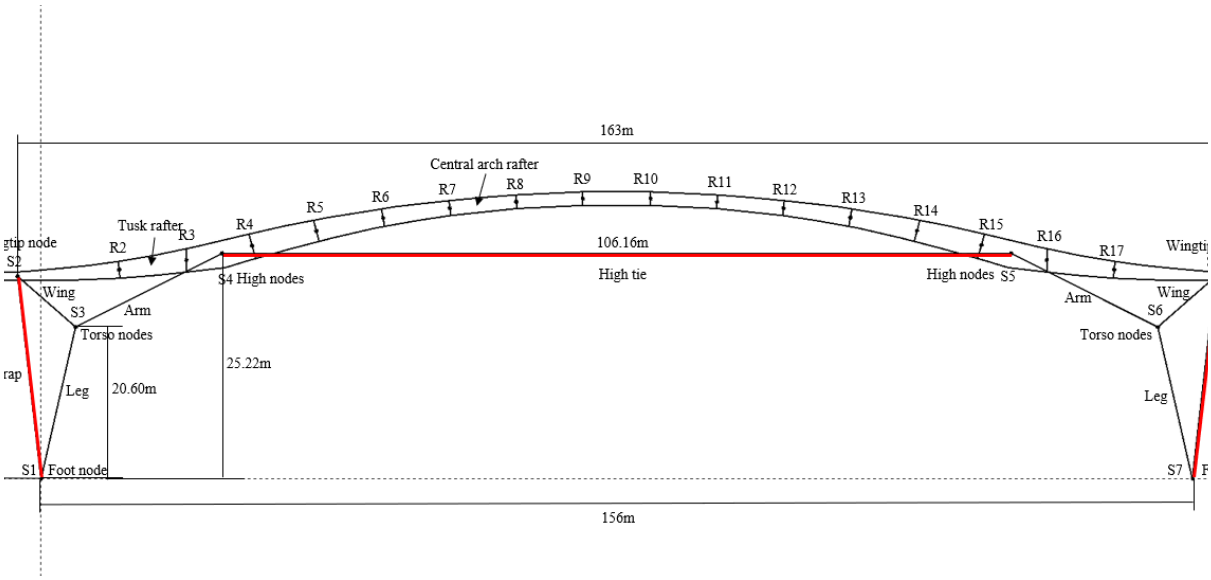
2 Deliverable 1: Geometry

The roof geometry was reconstructed from the supplementary drawing using a one-to-one SketchUp model and then transferred into GSA. As shown in Figure 1, points labelled R define the roof profile, while points labelled S define the support system and lower restraint points. The reconstructed model was checked against the original drawing on both sides of the roof and the overall geometric deviation was approximately 0.5%, which is acceptable for global FE modelling.

The second diagram shows the FE idealisation adopted for analysis. The curved roof was represented by short straight beam elements between adjacent roof nodes, while the high tie, straps, legs, arms and wings were represented as line members connected at the major nodes. This idealisation captures the main load path and preserves the key dimensions required for later loading, validation and mesh sensitivity studies.



(a) Reconstructed nodal geometry



(b) Idealised FE member layout

Figure 1: Geometry adopted for the global model.

3 Deliverable 2: Material, Section, and Loading Data

The global FE model was defined using linear-elastic steel. In accordance with the coursework brief, all members were assigned an elastic modulus of 200 GPa. A Poisson's ratio of 0.30 and density of $7850\,\mathrm{kg\,m^{-3}}$ were adopted so that self-weight could be included automatically in GSA. S355 steel was used for the primary roof members, which is consistent with the Heathrow T5 project information. The high tie was modelled with an initial prestress of 450 MPa.

Only vertical symmetric loading was considered. The characteristic permanent roof load was taken as the sum of the aluminium standing seam roof (0.10 kPa), structural deck (0.15 kPa), insulation/acoustic layers (0.40 kPa) and services allowance (0.15 kPa), giving $G_k = 0.80\,\mathrm{kPa}$. The characteristic imposed roof live load was taken as $Q_k = 0.60\,\mathrm{kPa}$. Therefore, the governing vertical ULS combination used in the model was

$$1.35G_k + 1.5Q_k = 1.35(0.80) + 1.5(0.60) = 1.98\,\mathrm{kPa}.$$

Using a tributary width of 36 m for one transverse frame, this corresponds to an equivalent applied line load of $71.28\,\mathrm{kN\,m^{-1}}$, in addition to the self-weight of the primary members.

The strap members were modelled as rod elements. Therefore, only their axial stiffness was represented in the global model, using an equivalent steel area of $A = 2 \times 500 \times 40 = 40\,000\,\mathrm{mm}^2$. The spacing between the two plates was neglected in this idealisation because rod elements do not carry bending.

Table 1: Section data adopted for the global model

Member	**Section**	**Modelling note**
Arch rafter	Welded box, 800 mm wide, 50 mm wall thickness	Depth varies from about 1.8 m to 3.8 m along the roof profile
Tusk rafter	Welded I-section, 50 mm plate thickness	Overall depth varies from about 0.9 m to 3.8 m
Legs and arms	CHS 914 mm × 40 mm	Taken directly from the supplementary sketch
Wings	CHS 711 mm × 50 mm	Taken directly from the supplementary sketch
Internal/external ties	CHS 457 mm × 40 mm	Thickness not stated in the source, so 40 mm was assumed
High tie	Twin locked-coil strands, 2 × 115 mm	Prestressed to 450 MPa
Straps	Equivalent rod area $40\,000\,\mathrm{mm}^2$	Based on two 500 mm × 40 mm plates; bending stiffness neglected

The main assumptions were therefore a linear-elastic S355 steel model, uniform wall thickness where required by the brief, a 36 m tributary width for each transverse frame, automatic inclusion of self-weight through the material density, and axial-only idealisation of the strap members.

4 Deliverable 3: Initial Model Validation

4.1 Assumptions

4.2 Model Validation

4.3 Model Adequacy

5 Deliverable 4: Mesh Refinement Sensitivity Analysis

5.1 Introduction and Methodology

In Finite Element Analysis (FEA), continuous curved geometries must be approximated using discrete straight beam elements. If the mesh is too coarse, the structural stiffness and internal bending moments of curved members are inaccurately represented. A sensitivity study was conducted to determine the optimum mesh density that provides mathematical convergence without introducing unnecessary computational expense.

The study selectively targeted the continuous curved roof profile, specifically the Arch and Tusk Rafters (including the cantilevered eaves). These members carry heavy distributed loads (72.74 kN/m) and are subjected to complex bending forces. Conversely, the straight high ties and the vertical abutment columns (legs and arms) were excluded from the refinement process; as primarily axially loaded members, their behaviour is perfectly captured by single elements between major nodes regardless of mesh density.

5.2 Mesh Definitions

Three levels of mesh refinement were analysed under the critical ULS vertical load combination ($1.35G_k + 1.5Q_k$) and the 450 MPa high tie prestress. The base geometry was defined by the 16 coordinate nodes provided in the project data.

The baseline geometry of Mesh 1 (coarse) consisting of 1 element per original coordinate segment across the roof profile. (Average arch element length $\approx$ 2.5 m). For Mesh 2 (medium), all elements along the continuous roof profile were split into two equal segments. (Average arch element length $\approx$ 1.25 m). For Mesh 3 (fine), all elements along the continuous roof profile were split again into two equal segments, resulting in four elements per original coordinate segment. (Average arch element length $\approx$ 0.6 m).

5.3 Comparison of Key Response Quantities

To evaluate convergence, three critical structural responses were monitored across the three models, maximum vertical deflection (δ_z), which was measured at the crown (apex) of the arch rafter, maximum bending moment (M_y) measured within the arch rafter, and maximum axial tension (F_x), which is measured in the main twin high tie.

Table 2: Mesh Sensitivity Results under ULS Loading

Parameter	**Mesh 1 (Coarse)**	**Mesh 2 (Medium)**	**Mesh 3 (Fine)**	**% Change (Med to Fine)**
Max Crown Deflection (mm)	[Output]	[Output]	[Output]	[%]
Max Arch Bending Moment (kNm)	[Output]	[Output]	[Output]	[%]
High Tie Axial Force (kN)	[Output]	[Output]	[Output]	[%]

5.4 Conclusion and Adopted Mesh Density

As demonstrated in Table 2, transitioning from the coarse mesh to the medium mesh resulted in noticeable changes to the internal forces and deflections. This indicates that the coarse mesh artificially bridges the true curvature of the arch, creating an overly stiff model that inaccurately distributes bending moments.

However, further refinement from the medium mesh to the fine mesh yielded a negligible variance of [%] in the key response quantities. This demonstrates that the FE results have converged and the global behaviour of the structure is no longer highly sensitive to element size.

Therefore, Mesh 2 (medium mesh) provides sufficient geometric accuracy for the purpose of global structural modelling. It captures the true bending behaviour of the sweeping arch rafters without introducing the unnecessary mathematical complexity of Mesh 3. Mesh 2 has been adopted for all subsequent global analyses in this report.

6 Deliverable 5: Influence of Prestressing Assumptions

6.1 Introduction and Modelling Basis

This section compares two prestressing assumptions for the main high tie under the same vertical loading. **Model A** applies prestress after full erection, following the coursework simplification. **Model B** represents the real construction sequence in a simplified way so that the influence of earlier stressing and load transfer can be assessed.

The aim of the comparison is to determine whether the simplified post-erection prestress assumption is sufficient for global analysis, or whether the construction-stage stressing process produces meaningful differences in internal force distribution, deflection and sensitivity to erection tolerances. Both models should therefore use the same geometry, mesh density and vertical loading so that the prestressing assumption is the only significant variable.

6.2 Comparison of Key Response Quantities

Table 3: Comparison of prestressing assumptions

Response quantity	**Model A**	**Model B**	**% Diff.**
Maximum crown deflection (mm)	[ ]	[ ]	[ ]
Maximum arch bending moment (kNm)	[ ]	[ ]	[ ]
Maximum high tie axial force (kN)	[ ]	[ ]	[ ]
Maximum leg/support force (kN)	[ ]	[ ]	[ ]
Support spread / horizontal movement (mm)	[ ]	[ ]	[ ]
Any other critical response	[ ]	[ ]	[ ]

Table 3 should be used to compare the two models quantitatively. If Model B gives lower arch bending moments and different locked-in tie forces, this would suggest that the construction-stage stressing process redistributes part of the self-weight response before the final structure is formed. If the differences are small, the simplified post-erection prestress assumption is likely to be acceptable for global analysis.

6.3 Discussion and Implications

The numerical comparison should then be supported by a qualitative discussion of the deformed shape and force flow. Particular attention should be paid to the crown deflection, the support spread, the force carried by the high tie, the support and leg response, and whether the arch becomes more or less bending-dominated under the two assumptions.

The supplementary information makes it clear that prestressing was integral to the erection sequence rather than a load applied only after the roof had been fully assembled. The centre arch was assembled and prestressed at low level, lifted into position, connected to the abutments and then subjected to staged load transfer, so the final structural behaviour depended on locked-in forces and redistribution developed during construction. This means that, if Model B shows meaningful differences in force or deflection, a staged-construction approach should be adopted for design because the simplified post-erection assumption may underestimate temporary compression effects and the resulting final force state. The same construction process also has important implications for tolerance control: the cables were stressed incrementally in pairs until the rafters lifted clear of their temporary supports, the geometry was monitored continuously during stressing, and final cable forces were adjusted again after creep. Tolerance was therefore achieved through active adjustment during erection as well as fabrication accuracy, so the design should provide sufficient allowance for cable tuning, splice fit-up and support positioning, together with preset geometry that can accommodate the larger movements that may occur during load transfer. For this reason, Model B should be preferred for the remainder of the report whenever it produces non-negligible differences, since it better reflects the documented construction method and the resulting locked-in structural response.

7 Deliverable 6: Boundary Conditions for More Detailed Analysis

For a more detailed FE study, the rafter splice between the tusk rafter and arch rafter should be extracted from the global frame model and analysed as a local submodel. The purpose of the local model would be to resolve the stress distribution in the splice plates, bolts or welds, and the adjacent box and I-section walls, which cannot be captured accurately by the beam idealisation used in the global model.

The local model should include a short length of member on each side of the splice so that the boundary conditions are applied away from the connection itself. This reduces artificial stress concentrations caused by restraining the model too close to the detail. A practical choice would be to include approximately one section depth of the arch rafter and one section depth of the tusk rafter beyond the splice on either side.

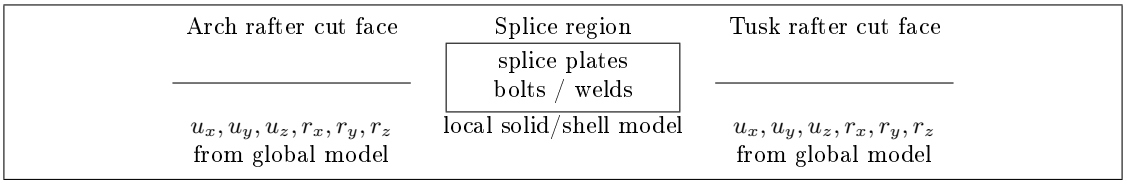


Figure 2: Recommended boundary conditions for the local splice model

The preferred boundary condition is to impose the displacement and rotation components extracted from the global beam model at the two cut faces of the local model. In practice, each cut face can be coupled to a reference node so that the sectional deformation remains compatible with the global model. The six kinematic quantities imposed at each end are the three translations (u_x, u_y, u_z) and the three rotations (r_x, r_y, r_z). This approach is preferable to simply fixing one end, because it reproduces the actual deformation state of the connection and avoids introducing artificial restraint moments.

The corresponding section forces should also be checked against the global model outputs. The main actions expected at the splice are axial force N, vertical shear V, bending moment M, and a smaller torsional component due to the curved roof geometry and eccentricity of the connection detail. If the software or modelling approach makes direct displacement transfer difficult, an acceptable alternative is to apply the equivalent set of sectional resultants at the two cut faces, provided that overall equilibrium is maintained.

To prevent rigid body motion in the local model, one reference node may be lightly stabilised after the global displacements have been assigned. This stabilisation should only remove numerical singularities and must not alter the transferred load path. Contact, bolt preload, or weld representation can then be added depending on the intended level of detail. For the purposes of this coursework, the key requirement is that the splice boundaries are driven by the global model response rather than by arbitrary fixed supports.